Experiment 9

Aim: Implement the Private Ethereum Blockchain using Geth.

Theory:

Introduction to Ethereum

Ethereum is a decentralized and open-source blockchain system that supports not only digital currency transactions but also programmable smart contracts and decentralized applications (DApps). It was introduced by Vitalik Buterin in 2013 and became operational in 2015.

While Bitcoin mainly serves as a digital currency, Ethereum extends this concept by acting as a distributed computing platform where developers can deploy automated programs known as smart contracts. Its native cryptocurrency, Ether (ETH), is used to pay transaction and computation fees, commonly called gas.

Ethereum maintains a shared global state that is continuously updated after every transaction or contract execution. Each node stores a copy of this data, ensuring decentralization and reliability. The platform follows an account-based structure, where accounts hold balances and interact with other accounts or smart contracts.

Private Ethereum Network

A **Private Ethereum Network** is a customized blockchain setup that operates separately from the public Ethereum mainnet while still following the same protocol. It is commonly used for development, testing, and academic purposes.

In such networks, parameters like consensus type, gas limits, block intervals, and account balances can be defined as needed. This flexibility allows developers to experiment and test applications without spending real cryptocurrency or exposing data publicly, making it ideal for controlled environments.

Geth (Go-Ethereum)

Geth is the official implementation of the Ethereum protocol developed in the Go programming language. It enables a machine to act as a full Ethereum node capable of processing transactions, executing smart contracts, and maintaining the blockchain.

Geth can be accessed via command-line tools, a JavaScript console, or APIs such as JSON-RPC. It supports both public networks and private blockchain setups, making it a commonly used tool for Ethereum development and deployment

Consensus Mechanism – Clique (Proof of Authority)

Private Ethereum setups often use the **Clique Proof of Authority (PoA)** consensus instead of Proof of Work. In this model, only selected and trusted nodes, called signers, are allowed to generate and validate blocks.

Since it does not require heavy computational work like mining, Clique offers faster transaction processing and lower energy consumption. The list of authorized signers is defined in the genesis block, making it suitable for private or consortium-based networks where participants are known.

Steps to Implement a Private Ethereum Network

1. **Install Geth:**
   Download and install the Geth client, and configure it so it can be accessed via the system command line.
2. **Create Accounts:**
   Generate Ethereum accounts for each node using Geth. These accounts include public addresses and encrypted private keys stored securely.
3. **Set Up Password Files:**
   Create password files to allow automatic unlocking of accounts when nodes start, avoiding repeated manual entry.
4. **Configure Genesis Block:**
   Define the initial blockchain configuration using a `genesis.json` file. This includes chain ID, consensus settings, gas limit, and initial balances.
5. **Initialize Nodes:**
   Initialize each node with the genesis file using the appropriate Geth command to prepare the blockchain environment.
6. **Start the Nodes:**
   Run the nodes, with one acting as the primary signer (block creator) and others participating as regular nodes.
7. **Connect Nodes:**
   Establish communication between nodes using enode URLs. Peer connections can be verified through Geth commands.
8. **Perform Transactions:**
   Execute transactions between accounts using Geth console commands. These transactions are validated, added to blocks, and reflected in account balances.

implementation:

Step 1: Install Geth

wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
mkdir -p ~/bin
cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113
export PATH="$HOME/bin:$PATH"
geth113 version

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
g2022_mahvish@mahvish@cloudshell:~$ wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
mkdir -p ~/bin
cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113
export PATH="$HOME/bin:$PATH"
geth113 version
--2026-04-09 15:28:07--  https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
Resolving gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)... 20.60.40.164
Connecting to gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)|20.60.40.164|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 28246613 (27M) [application/octet-stream]
Saving to: 'geth-linux-amd64-1.13.14-2bd6bd01.tar.gz'

geth-linux-amd64-1.13.14-2bd6bd01.tar.g 100%[===================================================================================>]  26.94M   765KB/s

2026-04-09 15:28:47 (715 KB/s) - 'geth-linux-amd64-1.13.14-2bd6bd01.tar.gz' saved [28248613/28248613]

geth-linux-amd64-1.13.14-2bd6bd01/
geth-linux-amd64-1.13.14-2bd6bd01/COPYING
geth-linux-amd64-1.13.14-2bd6bd01/geth
Geth
Version: 1.13.14-stable
Git Commit: 2bd6bd01d2e8551dd7fc21b631f4a34ac16627a1
Git Commit Date: 20240227
Architecture: amd64
Go Version: go1.21.6
Operating System: linux
GOPATH=/home/g2022_mahvish/gopath:google/gopath
GOROOT=
g2022_mahvish@cloudshell:~$
```

Step 2: Create folders and accounts for both nodes

```
g2022_mahvish@cloudshell:~ $ mkdir -p ~/eth-private/node1 ~/eth-private/node2
g2022_mahvish@cloudshell:~ $ geth113 --datadir ~/eth-private/node1 account new
INFO [04-09|15:36:17.038] Maximum peer count                       ETH=50 total=50
INFO [04-09|15:36:17.011] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm:  file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x93B0C547De2cF8228981Dd4a29aC3D0cA9a08859
Path of the secret key file: /home/g2022_mahvish/eth-private/node1/keystore/UTC--2026-04-09T15-36-35.166702641Z--93b0c547de2cf8228981dd4a29ac3d0ca9a0885
9

- You can share your public address with anyone. Others need it to interact wit yo.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access funds!
- You must REMEMBER YOUR password! Without the password, it's impossible to decrypt key!
```

```
g2022_mahvish@cloudshell:~$ geth113 --datadir ~/eth-private/node2 account new
INFO [04-09]15:38:49.547] Maximum peer count                       ETH=50 total=50
INFO [04-09]15:38:49.547] Smartcard socket not found, disabling    err="stat /run/pcscd/pscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:
Your new key was generated

Public address of the key:   0x6406586b898212f380A55Bbf5e6596bD95825D3b
Path of the secret key file: /home/g2022_mahvish/eth-private/node2/keystore/UTC--2026-04-09T15-38-54.774381074Z--6406586b898212f380a55bbf5e6596bd9
b

- You can share your public address with anyone. Others need it to interact you.
- You must NEVER share the secret key with anyone! The key controls to your funds!
- You must BACKUP your key file! Without the key, it's impossible to account funds!
- You must REMEMBER your password! Without the password, it's impossible to decypt the key!
```

Step 3: Create password files

```
g2022_mahvish@cloudshell:~$ echo "Mahvish" > ~/eth-private/node1/password.txt
g2022_mahvish@cloudshell:~$
```

Step 4: Create genesis.json and verify it looks correct

```
NODE1="PASTE_NODE1_ADDRESS_WITHOUT_0x_HERE"
NODE2="PASTE_NODE2_ADDRESS_WITHOUT_0x_HERE"
cat > ~/eth-private/genesis.json << EOF
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
      "period": 5,
      "epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "8000000",
  "extradata":
"0x0000000000000000000000000000000000000000000000000000000000000000${NODE1}0
000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000",
  "alloc": {
    "0x${NODE1}": { "balance": "100000000000000000000" },
    "0x${NODE2}": { "balance": "100000000000000000000" }
  }
}
EOF
```

```
> admin.nodeInfo.enode
"enode://acfe33406144ee52760abb42a72a912f3123188c228f201424f105b391186998f0fc78a77e454680d8898b9ca388cf647de953ae695bb21251e5f610baa5bb7d@34.126.99.110:30303"
> INFO [04-09|16:22:47.013] Successfully sealed new block      number=231 sealhash=bb4346..610566 hash=d1fc70..84fdc7 elapsed=5.001s
INFO [04-09|16:22:47.014] Commit new sealing work            number=232 sealhash=8aca46..08c3ca txs=0 gas=0 fees=0 elapsed="538.827µs"
INFO [04-09|16:22:52.010] Successfully sealed new block      number=232 sealhash=8aca46..08c3ca hash=d65637..b7846e elapsed=4.996s
INFO [04-09|16:22:52.010] Commit new sealing work            number=233 sealhash=e1df14..1d6155 txs=0 gas=0 fees=0 elapsed="485.675µs"
INFO [04-09|16:22:52.270] Looking for peers                  peercount=0 tried=60 static=0
INFO [04-09|16:22:57.011] Successfully sealed new block      number=233 sealhash=e1df14..1d6155 hash=0be857..1fcf8d elapsed=5.001s
INFO [04-09|16:22:57.012] Commit new sealing work            number=234 sealhash=51ca87..5b8c95 txs=0 gas=0 fees=0 elapsed="501.891µs"
```

```
g2022_mahvish@cloudshell:~$ NODE1="93B0C547De2cF8228981Dd4a29aC3D0ca9a08859"
NODE2="6406586b6892122380A55Bbf5e6596bd95825D3b"

cat > ~/eth-private/genesis.json << EOF
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eIp150Block": 0,
    "eIp155Block": 0,
    "eIp158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
      "period": 5,
      "Epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "8000000",
  "extradata": "0x0000000000000000000000000000000000000000000000000000000000000000$(NODE1)0000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000",
  "alloc": {
    "0x${NODE1}": { "balance": "10000000000000000000" },
    "0x${NODE2}": { "balance": "10000000000000000000" }
  }
}
EOF
```

```
g2022_mahvish@cloudshell:~$ NODE1="93B0C547De2cF8228981Dd4a29aC3D0ca9a08859"
NODE2="6406586b898212f380A55Bbf5e6596bD95825D3b"

cat > ~/eth-private/genesis.json << EOF
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eIp150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
      "period": 5,
      "Epoch": 30000
    }
  },
  "difficulty": "1",
  "gaslimit": "8000000",
  "extradata": "0x0000000000000000000000000000000000000000 0000000000000000093B0C547De2cF8228981Dd4a29ac3D0ca9a0885900000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000 0000000000000000000000000000",
  "alloc": {
    "0x93B0C547De2cF82289811Dd4a29aC3D0ca9a0859": { "balance": "1000000000000" },
    "0x6400586b898212f380A55Bbf5e6596bD95825D3b": { "balance": "1000000000000" }
  }
}
```

Step 5: Initialize both nodes

```
g2022_mahvish@cloudshell:~$ NODE1=NODE1="93B0c547De2cF8228981Dd4a29ac3D0ca9a08859"
NODE2="6405586bb8981223800AA55f5e6596bD95825D3b"
cat > ~/eth-private/genesis.json << EOF
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
  }, "clique": {
  "gasLimit": "${NODE1}",
  "extradata": "0x0000000000000000000000000000000000000000000000000000000000000000${NODE1}0000000000000000000000000000000000000000000000000000000000000000
0 "alloc": {
    "0x${NODE2}": { "balance": "10000000000000000000" }
EOF
g2022_mahvish@cloudshell:~$ geth113 --datadir -/eth-private/node2 init -/eth-private/genesis.json
INFO [04-09[15:49:52.113] Maximum peer peer                  ETH=50 total=50
INFO [04-09[15:49:52.111] Smartcard socket not disalng       scheme=hash
INFO [04-09[15:49:52.116] Intializing the  library           database=/home/g2022_mahvish/eth-private/node2/geth/chaindata cache=16.00MiB handles=
INFO [04-09[15:49:52.189] Allocated cache and file handles   database=/home/g2022_mahvish/eth-private/node2/geth/chaindata/ancient/chain readonly=
dles=16                   Opened ancient database            database=/home/g2022_mahvish/eth-private/node2/geth/lightchaindata cache=16.00MiB han
INFO [04-09[15:49:52.377] State schema set to default        database=/home/g2022_mahvish/eth-private/node2/geth/lightchaindata/ancient/chain read
INFO [04-09[15:49:52.373] Writing custom genesis block       scheme=hash
INFO [04-09[15:49:52.378] Persisted trie from memory database nodes=3 size=41.0B time=4.4763ms gnodes=0 gcsize=0.0B gctime=0s livenodes=0 livesize=0.
INFO [04-09[15:49:52.220] Successfully wrote genesis state   database=/chainadata hash=e03c67..887468
g2022_mahvish@cloudshell:~$
```

Step 6: Run Node 1 (signer)

```
cloudshell ×   cloudshell ×   +
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
g2022_mahvish@mahvish@cloudshell:~$ export PATH="$HOME/bin:$PATH"
NODE1="93B0C547De2cF8228981Dd4a29aC3D0cA9a08859"
g2022_mahvish@mahvish@cloudshell:~$
```

```
g2022_mahvish@cloudshell:~$ geth113 --datadir  ~/eth-private/node1 \
  --networkid 1234 \
  --port 30303 \
  --http --http.port 8545 \
  --unlock 0x${NODE1} \
  --password ~/eth-private/node1/password.txt \
  --mine \
  --miner.etherbase 0x${NODE1} \
  --allow-insecure-unlock \
  console
INFO [04-09|15:52:04.688] Maximum peer count                       ETH=50 total=50
INFO [04-09|15:52:04.691] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-09|15:52:04.694] Set global gas cap                       cap=50,000,000
INFO [04-09|15:52:04.695] Initializing the KZG library             backend=gcksg
INFO [04-09|15:52:04.767] Allocated trie memory caches             clean=154.60MiB dirty=256.00MiB
INFO [04-09|15:52:04.767] Using pebble as the backing database     database=/home/g2022_mahvish/eth-private/node1/geth/chaindata
INFO [04-09|15:52:04.767] Allocated cache and file handles         database=/home/g2022_mahvish/mahvish/eth-private/node1/geth/chaindata cache=512.00MiB handles
=524,288
INFO [04-09|15:52:04.797] Opened ancient database                  database=/home/g2022_mahvish/mahvish/eth-private/node1/geth/chaindata/ancient/chain readonly=
false
INFO [04-09|15:52:04.798] State scheme set to already existing     scheme=hash
INFO [04-09|15:52:04.799] Initialising Ethereum protocol           network=1234 dbversion=<nil>
INFO [04-09|15:52:04.803]
INFO [04-09|15:52:04.803] ------------------------------------------------------------------------------------------------------------
-----------------
INFO [04-09|15:52:04.803] Chain ID:  1234 (unknown)
INFO [04-09|15:52:04.803] Consensus: Clique (proof-of-authority)
INFO [04-09|15:52:04.803]
INFO [04-09|15:52:04.803] Pre-Merge hard forks (block based):
INFO [04-09|15:52:04.803]  - Homestead:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/h
omestead.md)
INFO [04-09|15:52:04.803]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/t
angerine-whistle.md)
INFO [04-09|15:52:04.803]  - Spurious Dragon/1 (EIP 155): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/s
purious-dragon.md)
INFO [04-09|15:52:04.803]  - Spurious Dragon/2 (EIP 158): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/s
```

```
INFO [04-09|15:52:04.971] Starting peer-to-peer node               instance=Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
INFO [04-09|15:52:04.991] New local node record                    seq=1,775,749,924,991 id=a9a288d6769625c8 ip=127.0.0.1 udp=30303 tcp=30303
INFO [04-09|15:52:04.994] IPC endpoint opened                      url=/home/g2022_prajjwal_pandey/eth-private/node1/geth.ipc
INFO [04-09|15:52:04.994] Started P2P networking                   self=enode://acfe33406144ee52760abb42a72a912f3123188c228f201424f105b391186998f0fc78a77e454680
d8898b9ca388cf647de953ae695bb21251e5f610baa5bb7d@127.0.0.1:30303
INFO [04-09|15:52:04.994] Generated JWT secret                     path=/home/g2022_prajjwal_pandey/eth-private/node1/geth/jwtsecret
INFO [04-09|15:52:04.995] HTTP server started                      endpoint=127.0.0.1:8545 auth=false prefix= cors= vhosts=localhost
INFO [04-09|15:52:04.996] WebSocket enabled                        url=ws://127.0.0.1:8551
INFO [04-09|15:52:04.996] HTTP server started                      endpoint=127.0.0.1:8551 auth=true  prefix= cors=localhost vhosts=localhost
INFO [04-09|15:52:06.133] Unlocked account                         address=0x93B0C547De2cF8228981Dd4a29aC3D0cA9a08859
INFO [04-09|15:52:06.133] Legacy pool tip threshold updated        tip=0
INFO [04-09|15:52:06.133] Legacy pool tip threshold updated        tip=1,000,000,000
INFO [04-09|15:52:06.133] Snap sync complete, auto disabling
INFO [04-09|15:52:06.134] Commit new sealing work                  number=1 sealhash=16e9bf..5ee3c4 txs=0 gas=0 fees=0 elapsed="200.259µs"
INFO [04-09|15:52:06.144] Successfully sealed new block            number=1 sealhash=16e9bf..5ee3c4 hash=6baf2d..432713 elapsed=10.718ms
INFO [04-09|15:52:06.145] Commit new sealing work                  number=2 sealhash=95fbef..db0a46 txs=0 gas=0 fees=0 elapsed="583.298µs"
INFO [04-09|15:52:06.149] Indexed transactions                     blocks=2 txs=0 tail=0 elapsed=3.670ms
Welcome to the Geth JavaScript console!

instance: Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
coinbase: 0x93b0c547de2cf8228981dd4a29ac3d0ca9a08859
at block: 1 (Thu Apr 09 2026 15:52:06 GMT+0000 (UTC))
 datadir: /home/g2022_prajjwal_pandey/eth-private/node1
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> INFO [04-09|15:52:07.394] New local node record                   seq=1,775,749,924,992 id=a9a288d6769625c8 ip=34.126.99.110 udp=30303 tcp=30303
INFO [04-09|15:52:11.010] Successfully sealed new block            number=2 sealhash=95fbef..db0a46 hash=4ad89a..cdb782 elapsed=4.864s
INFO [04-09|15:52:11.010] Commit new sealing work                  number=3 sealhash=6684cc..c8ae91 txs=0 gas=0 fees=0 elapsed="585.63µs"
INFO [04-09|15:52:15.529] Looking for peers                        peercount=0 tried=83 static=0
INFO [04-09|15:52:16.010] Successfully sealed new block            number=3 sealhash=6684cc..c8ae91 hash=d3d30d..b6a37c elapsed=4.999s
INFO [04-09|15:52:16.011] Commit new sealing work                  number=4 sealhash=2e769c..084439 txs=0 gas=0 fees=0 elapsed="672.511µs"
INFO [04-09|15:52:21.010] Successfully sealed new block            number=4 sealhash=2e769c..084439 hash=f585e4..392e05 elapsed=4.998s
INFO [04-09|15:52:21.010] Commit new sealing work                  number=5 sealhash=998266..47aa05 txs=0 gas=0 fees=0 elapsed="619.955µs"
INFO [04-09|15:52:25.681] Looking for peers                        peercount=0 tried=68 static=0
INFO [04-09|15:52:26.010] Successfully sealed new block            number=5 sealhash=998266..47aa05 hash=5865a2..bceb3e elapsed=4.999s
```

Step 7: Open new terminal tab and run Node 2 (member)

```
g2022_mahvish@cloudshell:~$ gethshell  export PATH="$HOME/bin:$PATH"
NODE2="6406586b898212f380A55Bbf5e6596bD95825D3b"
```

```
g2022_mahvish@cloudshell:~$ geth113  --datadir ~/eth-private/node2          --networkid 1234   --port 30304   --http --http.port 8546   --unlock 0x${NODE2}   --
password ~/eth-private/node2/password.txt   --allow-insecure-unlock   console
INFO [04-09|16:10:33.198] Maximum peer count                       ETH=50 total=50
INFO [04-09|16:10:33.199] Smartcard socket not found, disabling    err="stat /run/pscd/pscd.comm: no such file or directory"
INFO [04-09|16:10:33.195] Set global gas sap                       cap=50,000,000
INFO [04-09|16:10:33.193] Initializing the KZG library             backous=gokzg
INFO [04-09|16:10:33.268] Allocated crle memory caches             clear=154.00MiB dirty=256.00MiB
INFO [04-09|16:10:33.268] Using pebble so the backing database
INFO [04-09|16:10:33.268] Allocated cache and file handles         database=/home/g2022_mahvish/eth-private/node2/geth/chainta/chaindata cache=512.00MiB handles
=524,288
INFO [04-09|16:10:33.300] Opened ancient database                  database=/home/g2022_mahvish/eth-private/node2/geth/chainta/chaindata/ancient/chain readonly=
false
INFO [04-09|16:10:33.300] State scheme set to already existing     schema=hash
INFO [04-09|16:10:33.301] Initializing Ethereum protocol           network=1234 dbversion=8
INFO [04-09|16:10:33.302]
INFO [04-09|16:10:33.302] ----------------------------------------------------------------------------------------------------------------------------------
------------------
INFO [04-09|16:10:33.302] Chain ID:  1234 (unknown)
INFO [04-09|16:10:33.302] Consensus: Clique (proof-of-authority)
INFO [04-09|16:10:33.302]
INFO [04-09|16:10:33.302] Pre-Merge hard forks (block based):
INFO [04-09|16:10:33.302]  - Homestead:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/t
omestead.md)
INFO [04-09|16:10:33.302]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/t
angerine-whistle.md)
INFO [04-09|16:10:33.302]  - Spurious Dragon/1 (EIP 155): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/s
purious-dragon.md)
INFO [04-09|16:10:33.302]  - Spurious Dragon/2 (EIP 158): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/s
purious-dragon.md)
INFO [04-09|16:10:33.302]  - Byzantium:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/b
yzantium.md)
INFO [04-09|16:10:33.302]  - Constantinople:               #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/c
onstantinople.md)
INFO [04-09|16:10:33.302]  - Petersburg:                   #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/p
etersburg.md)
```

```
> admin.addPeer("enode://acfe33406144ee52760abb42a72a912f3123188c228f201424f105b391186998f0fc78a77e454680d8898b9ca388cf647de953ae695bb21251e5f610baa5bb7d@34.126
.99.110:30303")
true
> INFO [04-09|16:25:38.021] Looking for peers                        peercount=0 tried=45 static=1
```

```
INFO [04-09|16:10:33.302] Post-Merge hard forks (timestamp based):
INFO [04-09|16:10:33.302]
INFO [04-09|16:10:33.302] ----------------------------------------------------------------------------------------------------------------------------------
------------------
INFO [04-09|16:10:33.302]
INFO [04-09|16:10:33.302] Loaded most recent local block           number=0 hash=e03c67..887468 td=1 age=57y1mo2d
WARN [04-09|16:10:33.303] Loaded snapshot journal                  diffs=missing
INFO [04-09|16:10:33.303] Initialized transaction indexer          range="last 2350000 blocks"
INFO [04-09|16:10:33.303] Loaded local transaction journal         transactions=0 dropped=0
INFO [04-09|16:10:33.325] Enabled snap sync                        head=0 hash=e03c67..887468
INFO [04-09|16:10:33.326] Gasprice oracle is ignoring threshold set threshold=2
WARN [04-09|16:10:33.330] Unclean shutdown detected                booted=2026-04-09T16:01:48+0000 age=8m45s
WARN [04-09|16:10:33.331] Engine API enabled                       protocol=eth
WARN [04-09|16:10:33.331] Engine API started but chain not configured for merge yet
INFO [04-09|16:10:33.331] Starting peer-to-peer node               instance=Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
INFO [04-09|16:10:33.363] New local node record                    seq=1,775,750,508,266 id=40d680c8eeed08a8 ip=127.0.0.1 udp=30304 tcp=30304
INFO [04-09|16:10:33.366] Started P2P networking                   self=enode://39855208b63767867bd73a27f0fee7a0a210588ad82c875d858df2c94a59e2d547e132ef6deda610
1cfba712be53477af7154b6ab8e6a03b814bf90fcdd60526@127.0.0.1:30304
INFO [04-09|16:10:33.366] IPC endpoint opened                      url=/home/g2022_prajjwal_pandey/eth-private/node2/geth.ipc
INFO [04-09|16:10:33.366] Loaded JWT secret file                   path=/home/g2022_prajjwal_pandey/eth-private/node2/geth/jwtsecret crc32=0x10258536
INFO [04-09|16:10:33.368] HTTP server started                      endpoint=127.0.0.1:8546 auth=false prefix= cors= vhosts=localhost
INFO [04-09|16:10:33.369] WebSocket enabled                        url=ws://127.0.0.1:8551
INFO [04-09|16:10:33.369] HTTP server started                      endpoint=127.0.0.1:8551 auth=true  prefix= cors=localhost vhosts=localhost
INFO [04-09|16:10:34.473] Unlocked account                         address=0x6406586b898212f380A55Bbf5e6596bD95825D3b
WARN [04-09|16:10:34.526] Served eth_coinbase                      reqid=3 duration="30.392µs" err="etherbase must be explicitly specified"
Welcome to the Geth JavaScript console!

instance: Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
at block: 0 (Thu Jan 01 1970 00:00:00 GMT+0000 (UTC))
 datadir: /home/g2022_prajjwal_pandey/eth-private/node2
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> INFO [04-09|16:10:35.821] New local node record                    seq=1,775,750,508,267 id=40d680c8eeed08a8 ip=34.126.99.110 udp=30304 tcp=30304
INFO [04-09|16:10:43.413] Looking for peers                        peercount=0 tried=52 static=0
INFO [04-09|16:10:53.510] Looking for peers                        peercount=0 tried=29 static=0
```

Step 8: Connect the two nodes

In Node 1 console, get its enode

Now, switch to Node 2 console and paste the Node 1 enode

In Node 2 console, check they are connected

```
> admin.peers
[{
    caps: ["eth/68", "eth/69", "snap/1"],
    enode: "enode://00162513d2f3e10b7ccd0dc4a3d139842e3f9681147bf7367aa7fe40421ce3904f8a4ea0db1254cc2d0d8a0dfec08457c3938017bf424a952c92c70931183abb@47.187.149.99:31404",
    id: "26184e724beb8194376e59f1fd0493aee8d3cb40469857695c966147c8b8aeeb",
    name: "Nethermind/v1.36.0+31cb81b7/linux-x64/dotnet10.0.1",
    network: {
      inbound: false,
      localAddress: "10.88.0.4:55586",
      remoteAddress: "47.187.149.99:31404",
      static: false,
      trusted: false
    },
    protocols: {
      eth: "handshake",
      snap: "handshake"
    }
}]
```

Step 9: Send a transaction

In Node 1 console, send transaction to Node 2

```
> eth.sendTransaction({
......   from: eth.accounts[0],
......   to: "0x5fd63344c1f40712A9A85d9d7e2B304a6C558716",
......   value: web3.toWei(1, "ether")
...... })
INFO [04-05|13:43:48.276] Setting new local account                address=0x1D8b6d4F6C677D2fB66CA97FbD81bED2cD23c4D3
INFO [04-05|13:43:48.276] Submitted transaction                    hash=0x566ac1b0f280ebd7b6d4a68991416a36248ed3ec43bd24f8a08af6badab79089 from=0x1D8b6d4F6C677D2fB66CA97FbD81b
ED2cD23c4D3 nonce=0 recipient=0x5fd63344c1f40712A9A85d9d7e2B304a6C558716 value=1,000,000,000,000,000,000
"0x566ac1b0f280ebd7b6d4a68991416a36248ed3ec43bd24f8a08af6badab79089"
```

and check the balance of Node 2

```
> eth.getBalance("0x5fd63344c1f40712A9A85d9d7e2B304a6C558716")
1010000000000000000000
```

Conclusion:

This experiment demonstrated the setup of a private Ethereum network using the Geth tool. A two-node network was created by configuring the genesis block with the Clique Proof of Authority consensus, initializing both nodes, and connecting them using enode addresses.

After successful setup, the primary node generated blocks while both nodes communicated through peer connections. A transaction was carried out between the nodes and verified using balance checks, confirming proper functioning of the network.

Overall, the experiment helped in understanding private blockchain setup, node interaction, and transaction processing, while also showing the efficiency of Proof of Authority in terms of speed and lower computational requirements.